

10 PHOTOGRAPHY

Photography hasn't seen much action for a while. It's a numbers game with new DSLRs just having more megapixels and focus points, and slightly improved image processors. Nothing revolutionary. Then Lytro introduced the Light Field cinematography. Though their first two ventures didn't pan out well, Lytro Cinema, their upcoming video camera, is all set to revolutionise the movie industry.

The Lytro Cinema is a unique camera. It has a sensor capable of capturing 755 RAW megapixels, and the sensor itself is a foot and a half wide. Captured content can be rendered as either 4K or 2K videos running at



Lytro's 755 megapixel light field cinema camera

300 FPS. If all this still seems like just numbers, consider that Light Field Cinematography allows you to change focus, depth of field and shutter speed during post production. The camera records 3D depth of individual objects, and also stores angular data. What does this mean? You can remove objects from the scene, and don't need to reshoot them. You can do away with green screens altogether! Also, since angular data is stored, you end up with a virtualized holographic image. Because the camera creates models of every object it shoots, it includes positional data in 3D space of them. This camera will drastically reduce post production costs, reduce reshoots and save a ton of money on CG effects.

The next development is in sensors, which have thus far been flat. Various elements in the lens are what help change the field of view and focus, which means you have to have a separate module with

the sensor and then the lenses in front of it. Not anymore. Researchers at the Columbia University are working to build a flexible sheet camera. This isn't a new concept, but previous iterations suffered from non-fixable aliasing, because the lenses in front the sensor weren't flexible. With this new technology, both elements are flexible.

So how are these going to be put to use? You could wrap sheets around objects and see them from all directions. We couldn't think of a more apt example other than the all seeing eye from Lord of the Rings. Jokes aside, you can wrap vehicles with these sheets and improve driver assist systems.

11 GAMING

VR has melded with the gaming ecosystem and each platform has added a multitude of VR titles. The PlayStation VR has been quite successful on the console front while Oculus Rift and HTC Vive have led the charge for PC. Needless to say, VR on the PC is a much more fluid experience thanks to the abundance of processing power on hand. However, it still remains a niche as VR Head Mounted Displays still aren't cheap. Xbox on the other hand decided to sit this one out but Scorpio will change that as it's packing a lot more horsepower to give gamers a PC-like experience. However, there's one thing that's really going to drive AR/VR adoption, and that's Physically Collaborative Gaming (PCG).

Over the years multiplayer games have carved a bigger piece of the pie than single player games – the highest sales records are exclusively multiplayer titles. With PCG, gaming hardware will incorporate technologies such as Intel's RealSense cameras, which will digitise

human players, who will then take the role of your co-op partner. This is the closest you'll get to the "real thing".

A worrying trend is the faster hardware and software refresh rates. Every year we have a new CPU and GPU refresh, which previously took at least 2-3 years to happen. Software is updated way too often anyway, with something like Google's

Chrome browser releasing 10 or 15 major updates a year. This is OK for software though, because it doesn't cost the user anything more, but with hardware updating costs a bomb!

For consoles, we saw the Xbox One S and PlayStation Pro being announced around E3, about three years after the PS4 launched. But Sony's going to release the PlayStation Neo in 2017. That's in less than one year. The fact is, because console manufacturers having adopted the x86 architecture, game developers no longer have to develop for the PC separately. This has led to quite a lot of PC specific code making its way onto the console versions.

DLCs

<http://dgit.in/MxdRealityV>
<http://dgit.in/GameTrends>



Facebook's Oculus Rift gaming boot at Gamescom 2014

This means that where earlier developers used to have to cut code and optimise for restrictive console hardware, they can afford to be a little lazier now, and that opens up a new avenue for console manufacturers to make even more money. You can kiss your long support periods for consoles goodbye, because manufacturers will focus only on the latest few iterations of their hardware. In short, consoles are going the mobile way, and your only hope is to join the #PCMasterRace.

MORE MEGAPIXELS

<http://dgit.in/FlexCam>
<http://dgit.in/LytrOvid>
<http://dgit.in/SceneDetect>